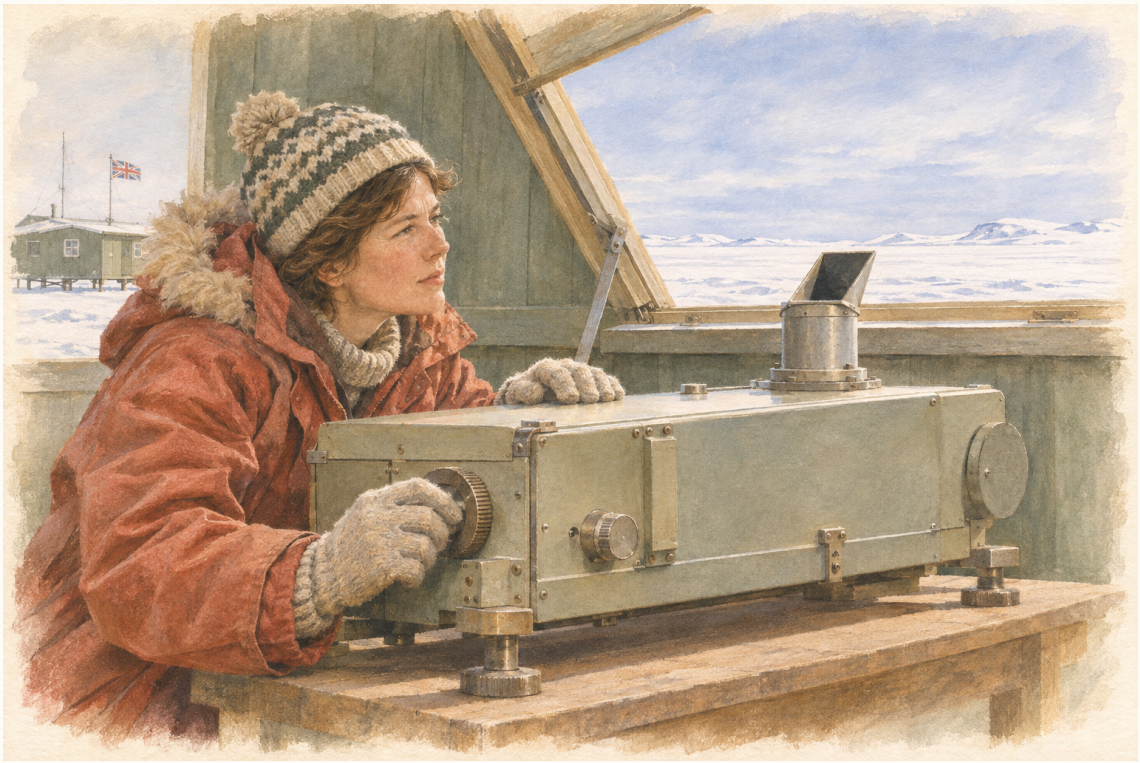
拯救天空

Saving the Sky



臭氧层的低值，为什么被软件怀疑?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

A layer of ozone shields Earth from much of the sun's ultraviolet radiation. In June 1974 Mario Molina and Sherwood Rowland, chemists in California, warned in Nature of a menace in fridges and spray cans. Chlorofluorocarbons (CFCs) are so inert that they drift intact into the stratosphere, high in the atmosphere, where they can liberate chlorine, a catalyst that destroys ozone molecule after molecule. The pair foresaw gradual erosion.

The alarm, when it sounded, was abrupt. In May 1985 Joseph Farman and colleagues at the British Antarctic Survey reported that ozone over Halley Bay, their Antarctic station, had plummeted each southern spring since the mid-1970s. Their humble Dobson spectrophotometer gauged ozone from the ground. The staff were persistent, reading it in a painstaking routine of manual observation.

Yet NASA's Nimbus-7 satellite, a sophisticated eye in orbit, had recorded the same lows. Its software flagged them as dubious, since ozone that low had never been reliably measured. NASA says it was still verifying them when Farman published. The sensor had worked; expectation had not.

Remarkably, the response was swift. In September 1987 diplomats signed the Montreal Protocol, a binding accord to curb CFCs. Amendments tightened it: in June 1990 affluent countries pledged to phase out CFCs by 2000. In September 2009 the Protocol and its parent Vienna Convention became the UN's first treaties to win universal ratification.

In January 2023 a UN-backed assessment forecast recovery to 1980 levels around 2066 over Antarctica, 2045 over the Arctic, and 2040 elsewhere. Healing will take decades. The sky's rescue began with diligent observers who trusted that a baffling number was no glitch, and published it.

中文译文

臭氧层为地球挡住了太阳的大量紫外线。1974年6月，在加州工作的化学家马里奥·莫利纳和舍伍德·罗兰在《自然》发出警告：冰箱和喷雾罐里藏着一种威胁。氯氟烃，也就是CFCs，性质很不活泼，能完整地飘入高空的平流层，在那里释放氯。氯是一种催化剂，会接连破坏臭氧分子。两人预料到的是逐步发生的损耗。

真正的警报却来得很突然。1985年5月，约瑟夫·法曼和英国南极调查局的同事报告：自20世纪70年代中期起，他们南极哈雷湾考察站上空的臭氧，每逢南半球春季都会骤降。他们使用一台不起眼的多布森分光光度计，从地面测量臭氧。工作人员持之以恒，遵循细致的固定流程，靠人工观测读数。

然而，美国国家航空航天局的雨云7号卫星，这只精密的太空眼睛，也已测到同样的低值。由于此前从未可靠地测到这么低的臭氧含量，软件把这些读数标成可疑。NASA表示，法曼发表论文时，他们仍在核实数据。传感器没有出错，出错的是人的预期。

各国的反应迅速得令人瞩目。1987年9月，外交官签署了具有约束力的《蒙特利尔议定书》，限制氯氟烃的使用。之后的修正案进一步收紧管控：1990年6月，富裕国家承诺在2000年前逐步淘汰氯氟烃。2009年9月，议定书及其所依据的《维也纳公约》，成为联合国首批获得所有国家批准的条约。

2023年1月，一份由联合国支持的评估报告预测，臭氧层恢复到1980年水平的时间，南极上空约为2066年，北极约为2045年，其他地区约为2040年。愈合仍需数十年。拯救天空的起点，是一群勤勉的观测者：他们相信，那组令人费解的数字不是故障，于是将其发表。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ozone n.
臭氧
layer of ozone shields Earth from much

menace n.
威胁
of a menace in fridges and spray cans.

inert adj.
化学性质不活泼的
inert that they drift intact into the

intact adj.
完好无损的
inert that they drift intact into the

liberate v.
释放
where they can liberate chlorine,

catalyst n.
催化剂
catalyst that destroys ozone molecule

erosion n.
侵蚀；逐步损耗
The pair foresaw gradual erosion.

abrupt adj.
突然的
alarm, when it sounded, was abrupt.

plummet v.
骤降
plummeted each southern spring since

persistent adj.
坚持不懈的
The staff were persistent, reading it

painstaking adj.
细致认真的
it in a painstaking routine of manual

sophisticated adj.
精密复杂的
a sophisticated eye in orbit,

dubious adj.
可疑的
Its software flagged them as dubious,

verify v.
核实
was still verifying them when Farman

binding adj.
有约束力的
a binding accord to curb CFCs.

accord n.
协定
a binding accord to curb CFCs.

curb v.
限制；遏制
a binding accord to curb CFCs.

pledge v.
承诺
June 1990 affluent countries pledged

phase out phr.v.
逐步淘汰
countries pledged to phase out CFCs by

ratification n.
正式批准
to win universal ratification.

assessment n.
评估
2023 a UN-backed assessment forecast

baffling adj.
令人费解的
a baffling number was no glitch, and

文中词汇与搭配

补充词汇共40项。保留每个表达在本文中的意思。

layer n.
层

shield from phr.v.
与from连用：保护；遮挡

from prep.
标示所防护的危害

ultraviolet adj.
紫外线的

radiation n.
辐射

spray n.
喷雾

atmosphere n.
大气层

chlorine n.
氯

molecule n.
分子

gradual adj.
逐渐的

alarm n.
警报

Antarctic adj.
南极的

humble adj.
不起眼的；简朴的

gauge v.
测量

routine n.
固定流程；例行程序

manual adj.
人工的；手工的

observation n.
观测

satellite n.
卫星

in orbit prep.phr.
在轨道上运行

flag as phr.v.
与as连用：标记为

文中词汇与搭配 续

补充词汇共40项。保留每个表达在本文中的意思。

as prep.
标示被认定的类别

sensor n.
传感器

expectation n.
预期

remarkably adv.
引人注目地

swift adj.
迅速的

diplomat n.
外交官

protocol n.
议定书

amendment n.
修正案

tighten v.
收紧；加强

affluent adj.
富裕的

treaty n.
条约

universal adj.
全体的；覆盖所有国家的

forecast v.
预测

recovery n.
恢复

Antarctica proper n.
南极洲

Arctic proper n.
北极地区

healing n.
愈合；修复

rescue n.
拯救

diligent adj.
勤勉的

glitch n.
小故障

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a _____ that destroys ozone molecule after molecule.

02 Its software flagged them as _____,

03 a _____ accord to curb CFCs.

04 affluent countries pledged to _____ CFCs by 2000.

文章主线

- 预警 臭氧挡紫外线。1974年，化学家警告：冰箱和喷罐的氯氟烃可进入平流层，释放氯持续破坏臭氧。
- 发现 1985年南极地面观测报告：哈雷湾臭氧自70年代中期起，每逢南半球春季便骤降。
- 误判 卫星同样测到低值，软件因罕见而标为可疑；NASA称当时尚在核实，是预期出了错，并非仪器失灵。
- 合作 各国1987年签署《蒙特利尔议定书》，1990年富国承诺2000年前淘汰氯氟烃，议定书2009年获所有国家批准。
- 恢复 2023年评估预测恢复至1980年水平：其他地区约2040年、北极约2045年、南极约2066年。

参考答案　1 catalyst 2 dubious 3 binding 4 phase out